

Week 1

Today is about forming a group for team-based learning and for group work.

Team name: Due tomorrow

Team member: Kla

: Bernedict

: Conner

: Brent

preferred communication platform: Discord

Team expectation: Do assigned work early

: Notify others if late

: Try responding in timely manner

Team rules : If team member doesn't do their work or doesn't communicate
potential grounds for exclusion.

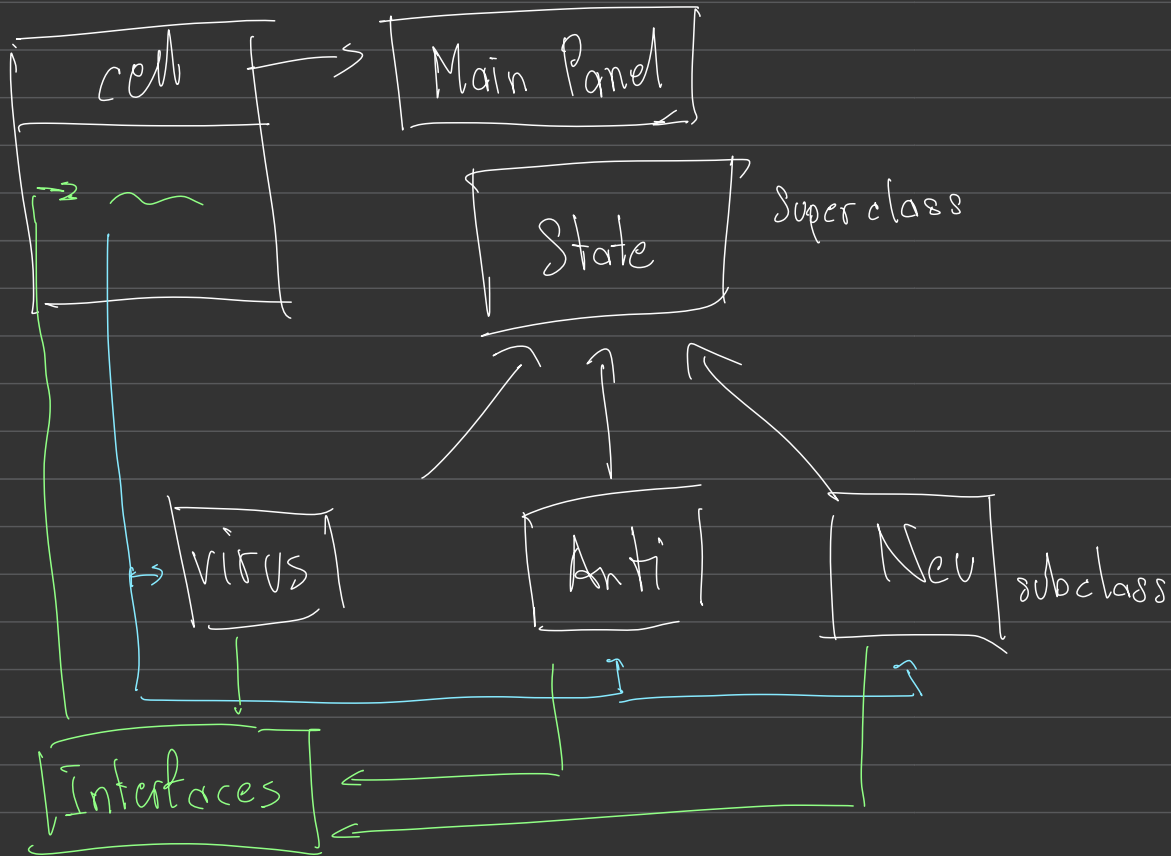
My potential concern is, mark is lower than expectation as I'm aiming for high score to keep my scholarship but marks are given as group work so it's harder to control and manage.

Week 1

- Learning Git and git commands
 - push/pull
 - commit
 - add
 - log - (using log -... to search history)
- Commit messages / Commit history
- Most creative approach from another team is used "--allow-empty commits"
So it is not necessary to make any changes
- Using log history to communicate
- Various ways to write git commits message
 - multiline
 - Attachment

Week 4

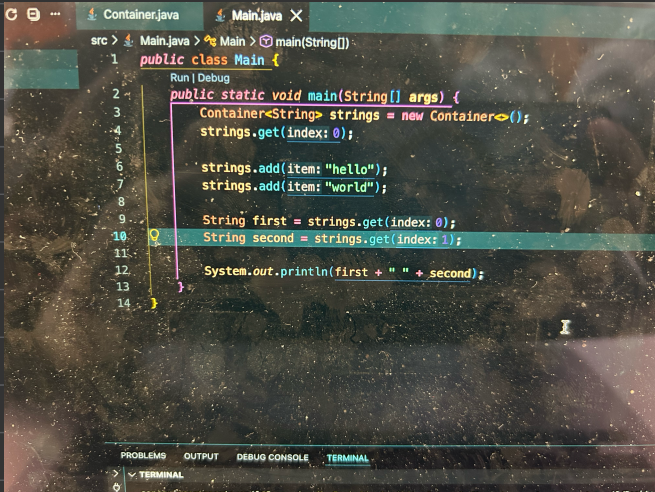
This UML diagram to show inheritance



Week 5

Modify the code in the given files (Main.java, container.java) to cause runtime exception without using casting

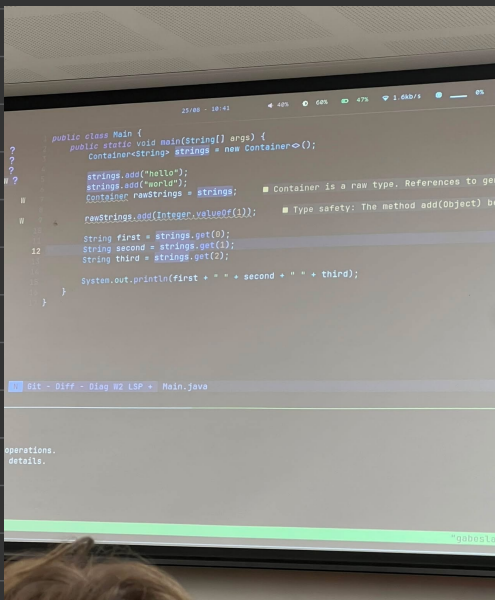
My solution



```
src > Main.java > Main > main(String[])
1 public class Main {
2     public static void main(String[] args) {
3         Container<String> strings = new Container<>();
4         strings.get(index: 0);
5
6         strings.add(item: "hello");
7         strings.add(item: "world");
8
9         String first = strings.get(index: 0);
10        String second = strings.get(index: 1);
11
12        System.out.println(first + " " + second);
13    }
14 }
```

→ Use index out of bounds

favourite solution



```
public class Main {
    public static void main(String[] args) {
        Container<String> strings = new Container<>();

        strings.add("hello");
        strings.add("world");
        Container rawStrings = strings;
        rawStrings.add(Integer.valueOf(1));

        String first = strings.get(0);
        String second = strings.get(1);
        String third = strings.get(2);

        System.out.println(first + " " + second + " " + third);
    }
}
```

→ Create raw type Container which is a reference copy of String container

→ Add int into raw type container

→ bypass compiler because raw container doesn't check type

→ Retrieve int from string container

→ Error

Week 6

Solving target barcode

Our team can solve up to level 6 but we got it wrong on level 4

Code for level 6

```
public class main {  
    public class BarcodePuzzle {  
        public static void main(String[] args) {  
            try {  
                star();  
            } catch (RuntimeException e) {  
            }  
  
            System.out.println();  
        }  
  
        static void star() {  
            System.out.print(s: "* ");  
            pipe();  
            pipe();  
  
            System.out.print(s: "* ");  
        }  
  
        static void pipe() {  
            System.out.print(s: "| ");  
            caret();  
            System.out.print(s: "| ");  
        }  
  
        static void caret() {  
            System.out.print(s: "^ ");  
            chuck_a_fit();  
            System.out.print(s: "^ ");  
        }  
  
        static void chuck_a_fit() {  
            iteration++;  
            // add your throws here  
            if (iteration == 2) {  
                throw new RuntimeException();  
            }  
        }  
    }  
}
```

→ catch exception in main function so the code skip the second "*" print

```
BarcodePuzzle.java  
File Edit View  
System.out.println();  
}  
  
static void star() {  
    System.out.print("* ");  
    try {  
        pipe();  
    } catch (Exception e) {  
    }  
    System.out.print("* ");  
}  
  
static void pipe() {  
    System.out.print("| ");  
    try {  
        caret();  
    } finally {  
        System.out.print("| ");  
    }  
}  
  
static void caret() {  
    System.out.print("^ ");  
    chuck_a_fit();  
    System.out.print("^ ");  
}  
  
static void chuck_a_fit() {  
    throw new RuntimeException();  
}  
}
```

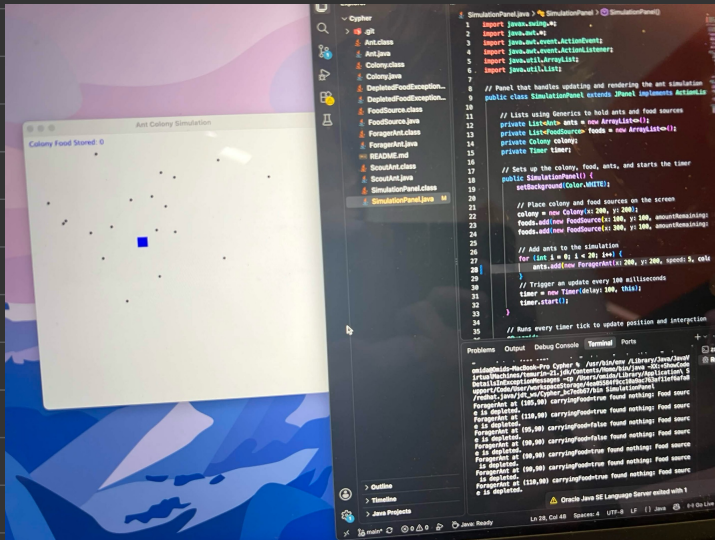
Most creative solution and correct solution for level 4

→ try-finally in one function

→ try-catch in another function

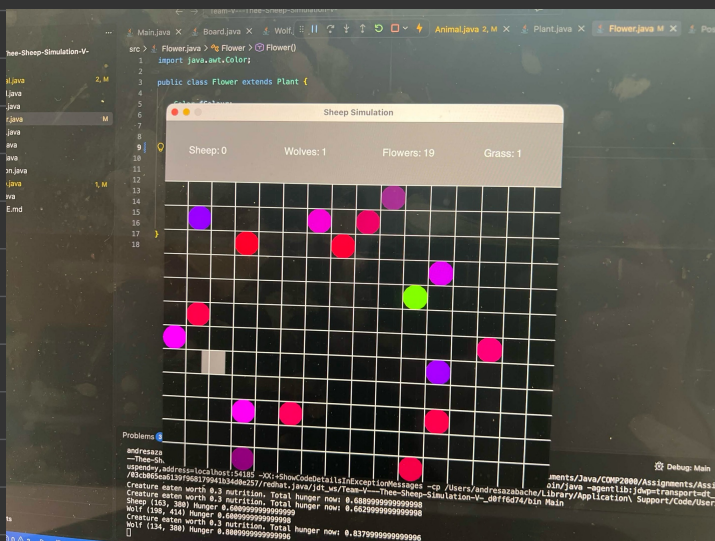
Week 7

One of the group did ant colony, where ants go randomly around the simulation panel and once it finds food, it will come back to its colony



- Small dots represent ants
- The square in the middle represents the center of its colony

Another group presented a wolf, sheep hunting grid simulation if the sheep and are in the same grid the sheep will be eaten



- Red and purple dots represent flower
- Green represents grass area
- Gray square represent the wolf

